

# Conditional Branching

→ `int main() {`



`return 0;`  
`}`

`int main() {`

`int marks;`

`printf("Enter marks: ");`

`scanf("%d", &marks);`

`if (marks >= 40) {`

`printf("passed");`

`printf("\ncongrats");`

`}`

`else {`

`printf("failed");`

`printf("\nBetter luck next time");`

`}`

`printf("\nthanks!");`

`return 0;`

`}`

|    |       |
|----|-------|
|    |       |
| 36 | marks |
|    |       |

# ★ Nested if

```
int main() {
    int marks;
    scanf("%d", &marks);
```

(28)

< 40

if  $\geq 35$

↑

35 → 40

36 → 40

37 → 40

38 → 40

39 → 40

```
    if (marks < 40) {
        if (marks  $\geq$  35) {
            marks = 40;
            printf("passed with grace");
        }
        else
            printf("failed");
        printf("Better luck next time");
    }
    else
        printf("passed");
    return 0;
}
```

①  $\geq 40$   
→ passed

②  $< 40$   
↳ ②.1  $\geq 35$   
↳ passed with grace  
②.2  $< 35$   
failed

X [ else  
printf("passed");  
return 0;  
}

|                                                              | I/p | O/p                 |
|--------------------------------------------------------------|-----|---------------------|
| int marks;<br>scanf("%d", &marks);                           | 95  | great work!<br>nice |
| [ if(marks > 90)<br>printf("great work!");                   | 85  | nice                |
| [ if(marks > 70)<br>printf("nice");<br>else<br>printf("ok"); | 65  | ok                  |

---

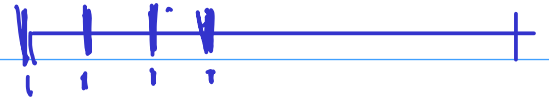
if( \_\_\_\_\_ ) {

\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_

}

dangling  
else [ else {  
          }

```
int marks;
scanf("%d", &marks);
```



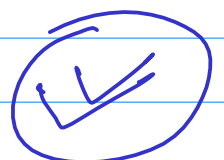
```
if (marks <= 100 && marks >= 91)
    printf("A");
if (marks <= 90 && marks >= 81)
    printf("B");
if (marks <= 80 && marks >= 71)
    printf("C");
if (marks <= 70 && marks >= 61)
    printf("D");
else
    printf("E");
```

|        |     |
|--------|-----|
| 91-100 | → A |
| 81-90  | → B |
| 71-80  | → C |
| 61-70  | → D |
| 60-0   | → E |

95-100 → A  
90-94 → B

```
if (marks >= 91)
    printf("A");
else
    if (marks >= 81)
        printf("B");
    else
        if (marks >= 71)
            printf("C");
        else
            if (marks >= 61)
                printf("D");
            else
                printf("E");
```

```
if (marks >= 91)
    printf("A");
else if (marks >= 81)
    printf("B");
else if (marks >= 71)
    printf("C");
else if (marks >= 61)
    printf("D");
else
    printf("E");
```



Ladder

if else

```

char fruit;
scanf("%c", &fruit);
if ('M' == fruit) {
    printf("I am Mango");
    printf("\n I am king of fruits");
}
else if ('O' == fruit) {
    printf("I am orange");
    printf("\n I am orange");
}
else if ('P' == fruit) {
    printf("I am pineapple");
    printf("\n I am sour and sweet");
}
else if ('W' == fruit) {
    printf("I am water melon");
    printf("\n I am watery");
}
else {
    printf("I do not know fruit
    starting with %c", fruit);
}

```

|     |       |                    |
|-----|-------|--------------------|
| 'M' | fruit | M - mango          |
|     |       | O - orange         |
|     |       | P - pineapple      |
|     |       | W - water<br>melon |

```
char fruit;  
scanf("%c", &fruit);
```

```
switch( fruit ) {
```

```
    case 'm':
```

```
        printf("I am mango");
```

```
        printf("\n I am hero");
```

```
        → break; ↘
```

```
    case 'o':
```

```
        printf("I am orange");
```

```
        → break; ↘
```

```
    case 'p':
```

```
        printf("I am pineapple");
```

```
        → break; ↘
```

```
    default:
```

```
        printf("No fruit starting from %c", fruit);
```

```
}
```